Session 2 (FIND & READ): Solutions and facilitation notes

Foundations of Clinical Research · Level 1 · Neudata · *Instructor copy. Do not share before the session.*

Model answers. Other answers are fine if they are specific and consistent.

Recap quiz: Session 1 answers

1. **Paper 1 (Boulet 2024) is a trial** (pilot RCT); **Paper 2 (Hyun 2023) is a cohort** (matched registry cohort).
2. **P** adults with sepsis · **E** positive cumulative fluid balance by day 3 · **C** neutral or negative balance · **O** death · **T** by day 28 (a PECO question: E = Exposure).
3. **Feasible**: a mortality outcome needs more than 1,000 patients.
4. **"To estimate the proportion fluid-positive on day 3"**: a measurable verb. "Understand" is not measurable.
5. **H0:** cumulative fluid balance at day 5 is the same with the restrictive protocol and with usual care.

If many miss 4 or 5, spend one extra minute: objectives use measurable verbs; H0 always states "no difference".

Check your understanding (segment 2.1): AND, OR or NOT?

1. **AND** (septic shock AND fluid restriction). 2. **OR** (restrictive OR conservative). 3. **Fewer** (AND narrows). 4. It removes every paper that mentions children anywhere, including good adult studies that mention children once.

Check your understanding (segment 2.2): OK or not OK?

1. **Not OK**: patchwriting is plagiarism even with a citation. 2. **OK** (but rare in medical writing). 3. **Not OK**: someone else's finding needs a citation. 4. **Not OK**: you may repeat an error, so read and cite the original.

Check your understanding (segment 2.3): spot the spin

1. Causal word for an observational design. Rewrite as *"a positive fluid balance was associated with higher mortality"*.
2. p = 0.31 is no evidence of a difference. Rewrite as *"no significant difference; the confidence interval was wide"*.
3. Headline is a secondary outcome. Lead with the primary result instead.
4. Before–after change without a comparison group. Rewrite as *"mortality was lower after implementation; without a control group we cannot attribute this to the protocol"*.

Exercise 1: Boolean search builder

Warm-up: `septic shock OR fluids` gives more hits (either term). `sepsis` gives more hits than `sepsis NOT neonatal` (NOT removes papers).

Model string for the running example (as in the demo):

```
("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab])
AND ("Fluid Therapy"[Mesh] OR "fluid restrict*"[tiab] OR "restrictive fluid*"[tiab] OR "conservative
AND ("Mortality"[Mesh] OR mortality[tiab] OR death[tiab])
```

Sensible extra synonyms for the I block: `de-resuscitation[tiab]`, `deresuscitation[tiab]`, `"fluid balance"[tiab]`, `"Water-Electrolyte Balance"[Mesh]`.

Sensible filters (each with a reason): article type *Randomized Controlled Trial* + *Systematic Review* (treatment question); *Adult: 19+ years* (our population); publication date *10 years* (current practice). Avoid *Free full text* (loses key papers).

Model string for Scenario C (delirium)

```
(Delirium[Mesh] OR delirium[tiab] OR "acute confusion"[tiab])
AND ("Hip Fractures"[Mesh] OR "hip fracture*"[tiab])
AND (prevent*[tiab] OR "multicomponent"[tiab] OR "non-pharmacological"[tiab] OR bundle[tiab])
```

Common faults

- Missing brackets: `A OR B AND C` is read differently from `(A OR B) AND C`.
- AND inside a block (`septic shock AND sepsis`). Synonyms need OR.
- A block for the comparison ("usual care"). It adds nothing and loses papers.
- `NOT` used to remove a topic the paper only mentions in passing.
- Search words in the local language: PubMed indexes English titles and abstracts.

Exercise 2: Paraphrasing (model answers)

Part A

Version	Verdict	Why
a)	**Problem: patchwriting**	Same structure and almost the same words as the source; one word swapped ("mortality" for "death"). Plagiarism even though it is cited.
b)	**OK**	New structure, adds context (matched registry cohort, 28-day mortality), keeps the calibrated word "associated", gives numbers and the citation.
c)	**Problem: distorts the source**	"Giving more fluid ... doubles the risk" turns an observational *association* into causation, and "doubles" overstates (HR 2.13 only for day 2; 1.56 for day 3). Citation does not fix misrepresentation.

Version	Verdict	Why
d)	**Problem: no citation**	Good wording, but presenting someone else's finding without a citation is plagiarism.

Part B: model paraphrase

"Mortality and days alive without organ support at day 28 did not differ between the two groups, although the pilot trial was too small to detect such differences [1]."

What good looks like: new structure; facts unchanged (28 days, mortality, days alive without life support, similar); no stronger claim (do **not** write "restriction is safe"); citation. Adding "the trial was too small" is accurate context from the paper (48 patients; not powered for mortality).

Exercise 3: Paper 1's abstract (answer key)

Item	Answer (from the abstract)
Design (and blinding)	Multicentre, randomised, controlled **pilot** trial; single-blind (patients and statisticians blinded, clinicians not)
P	Adults 18–85 years with septic shock, in the first 24 h of vasopressor infusion, in three ICUs; excluded: AKI needing renal replacement, end-stage CKD, severe malnutrition
I vs C	Optimised fluid restriction (reducing maintenance fluids and fluids for drug dilution during the first 7 days) vs standard fluid strategy
Primary outcome	**Cumulative fluid balance (ml/kg) in the first 5 days**, not mortality
Numbers	1,201 screened; 50 randomised; 2 withdrew (control); 48 analysed (24 per group)
Main result	Median cumulative balance at day 5: 6.9 vs 35.0 ml/kg; absolute difference 13.2 ml/kg (95% CI −15.2 to 41.6), p = 0.42. The 13.2 is a difference in means (the Methods report the mean difference with its 95% CI), so it does not equal the gap between the medians.
Conclusion	"...did not reduce the overall fluid balance at day 5"
Matches the result?	**Yes**. The CI includes zero, and the conclusion says "did not reduce".
Spin in the abstract?	**No** in the abstract conclusion. (Spin appears in the *Discussion*, which calls p = 0.42 a "non-significant trend".) Watch-point: the fluid *volumes* (89.7 vs 114.3 ml/kg, p = 0.0506) are a secondary result, and the abstract reports four decimals for a borderline p. This is mild emphasis, not a claim.
Registration	ClinicalTrials.gov **NCT04947904** (registered 1 July 2021, before recruitment began in September 2021)
Answers our 28-day mortality question?	**No.** It is a small pilot powered for fluid balance. 28-day mortality was a secondary outcome and was similar between groups: "no evidence of a difference", not "no difference".

Fast finishers (Paper 2): the title says **"Impact"** (causal); the abstract conclusion correctly says **"associated"**.

Common errors to correct in the debrief

- Writing "mortality" as the primary outcome.
- Copying only the p-value and not the estimate with its 95% CI.
- Calling the trial "double-blind". Clinicians, who prescribe the fluids, were not blinded.
- Reading the medians (6.9 vs 35.0) as "five times less fluid balance, so it works". The CI crosses zero.

Exercise 4: Proposal builder 2 (what good looks like)

The trainer models this live for the running example (the feasible pilot version A from Session 1); each participant then does the same for their own question. Use it as the quality benchmark, not as the answer.

Search (documented): PubMed, date, the string from Exercise 1, filters (article type RCT + systematic review; adults; 10 years), number of hits; plus a check of ClinicalTrials.gov for ongoing trials.

Working title: *Restrictive versus usual fluids in adults with septic shock: a pilot randomised trial* (13 words).

Keywords: Fluid Therapy; Water-Electrolyte Balance; Critical Care; Vasoconstrictor Agents (MeSH terms not already in the title).

Problem statement (≈ 300 words, three paragraphs)

1. *Known:* septic shock is a common cause of ICU death. Early fluid resuscitation restores perfusion, but continued fluid often produces a positive fluid balance, which observational studies associate with higher mortality (e.g. Hyun 2023).
2. *Not known:* large trials of restrictive fluids (CLASSIC 2022; CLOVERS 2023) found no mortality difference, and a pilot targeting hidden "non-resuscitative" fluids (Boulet 2024) did not significantly reduce fluid balance at day 5. Whether a restrictive protocol reliably reduces fluid balance is still unclear.
3. *Question and objectives:* in adults with septic shock, does a restrictive fluid protocol, compared with usual care, reduce cumulative fluid balance at day 5? Primary objective: to compare day-5 fluid balance between the two groups.

Science rule: the Introduction never contains your own data or results (no "our audit"). They belong in the Results. A local figure may appear only if it comes from a publication, with its reference.

Five key references (model)

#	Reference	Why included
1	Boulet N, et al. *Crit Care* 2024;28:429 (pilot RCT, n = 48)	Closest design and outcome; primary outcome not significantly different
2	Hyun DG, et al. *Ann Intensive Care* 2023;13:105 (matched cohort)	Positive balance on days 2–3 associated with higher 28-day mortality

#	Reference	Why included
3	Meyhoff TS, et al. (CLASSIC) *N Engl J Med* 2022;386:2459–70	Large RCT, septic shock, no difference in 90-day mortality
4	Shapiro NI, et al. (CLOVERS) *N Engl J Med* 2023;388:499–510	Large RCT, sepsis-induced hypotension, restrictive vs liberal, no difference in 90-day death before discharge
5	Evans L, et al. Surviving Sepsis Campaign guidelines 2021. *Crit Care Med* 2021;49:e1063–e1143	Current guideline; shows the remaining uncertainty

Instructor note: check citation details against PubMed before distributing. Participants must verify every reference themselves, which is exactly the habit to practise.

Feedback guide when reading drafts (not a grade)

Element	Good	Needs work
Search	String, database, date, filters and hits recorded; brackets correct	"We googled it"; no date; AND inside a block
References	5, relevant, include the best available design, exported to a reference manager	Websites, textbooks only, typed by hand, unread
Literature review worksheet	Same columns for every paper; limitation column filled	Copied abstract sentences
Problem statement	Known, then gap, then question and objectives; each fact cited	A literature dump; own data or results; unpublished local figures
Title & keywords	10–16 words; P, I/E, C and design visible; calibrated words; keywords add MeSH not in the title	"Fluids in the ICU"; abbreviations; "impact of" for an observational design
Integrity	Own words, every claim cited	Sentences pasted from abstracts

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Exit question: answers

1. **OR** gives more hits.
2. Paper 1's primary outcome was **cumulative fluid balance at day 5 (ml/kg)**, not mortality.